Anindita Ghosh

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Education

International Max Planck Research School on Trustworthy Computing — October 2021 – Present
Ph.D. in Computer Science — *Advisors: Dr. Philipp Slusallek, Dr. Christian Theobalt* — *Saarbrücken, Germany*

Saarland University — 2018 – 2021
M.S. in Visual Computing — *Advisor: Dr. Philipp Slusallek* — *ECTS Grade: 1.5* — *Saarbrücken, Germany*

St. Xavier's College — 2014 – 2016
M.Sc. (Hons.) in Computer Science — *Advisor: Dr. Asoke Nath* — *GPA: 8.31/10.00* — *Kolkata, India*

St. Xavier's College — 2011 – 2014
Bachelor of Science in Computer Science — *GPA: 7.81/10.00* — *Kolkata, India*

Professional Experience

Researcher — August 2021 – Present
DFKI Saarbrücken, Germany — *Supervisor: Dr. Philipp Slusallek, Team Leader: Dr. Klaus Fischer*

- Working with the Agents and Simulated Reality group on human motion analysis, modelling, and synthesis.

Research Assistant — May 2019 – July 2021
DFKI Saarbrücken, Germany — *Advisors: Erik Herrmann, Noshaba Cheema*

- Worked with the Agents and Simulated Reality group on parts of natural human motion simulation software and deep learning based text to motion synthesis.

Teaching Assistant — April 2020 – September 2020
Saarland Informatics Campus, Saarbrücken, Germany — *Professor: Dr. Joachim Weickert*

- Delivered tutorials, graded assignments and exams for a graduate-level course on Image Processing and Computer Vision.

Visiting Scholar — February 2018 – July 2018
Indian Statistical Institute, Kolkata, India — *Supervisor: Dr. Dipti Prasad Mukherjee*

- Worked on estimating the constituents of coal samples from the microscopic petrographic images, using image processing and machine learning techniques.

Research Associate — March 2017 – November 2017
Indian Institute of Science, Bengaluru, India — *Supervisor: Dr. Gurunath Gurrala*

- Developed an adaptive algorithm that can accurately estimate frequency and phase of grid signals in real-time at a high sampling rate.

Software and Programming Skills

ML and Vision	Python (PyTorch, Tensorflow), MATLAB, C++
Web Design	Angular JS, HTML5, Markdown, CSS, JavaScript
Graphics	OpenGL, WebGL, Unity + C#, Blender + Python
Documentation	LaTeX

Languages

Fluent	English, Bengali, Hindi
Beginner	German

Projects

Synthesis of Compositional Animations from Textual Descriptions *(M.S. thesis, Saarland University)*

- *Aim:* To generate an appropriate sequence of 3D human poses corresponding to complex natural language text input describing single or simultaneous actions.
- Designed a two-stream hierarchical sequence-to-sequence network that learns a joint embedding of language and poses of the upper and the lower bodies.
- Advanced the state-of-the-art on text-based motion synthesis in both subjective and objective evaluations.

MOSIM: End-to-end digital integration based on modular simulation of natural human motions *(DFKI)*

- Extended a motion capture web viewer based on Unity by adding the support of importing digital character models consisting of a skeleton and a textured mesh at runtime.
- Extended the REST API of the back-end of the server written in Python with functions to store and delete the data.
- Extended an Angular JS application that embeds the web viewer to prevent unauthorized users from access to this functionality.

UPLINX: Cross-location machine learning qualification program for practice *(DFKI)*

- Provided support for experiments with reinforcement learning for imitating motion capture by developing an importer and exporter for the custom motion format of a state of the art approach.

Sem2Pencil: From Segmentation Maps to Pencil Sketch Illustrations *(course project, Saarland university)*

- Modified the Spatially Adaptive Normalization technique of NVIDIA to generate pencil sketch illustrations from segmentation mappings for landscape images.

E-sense: Sensing and Analytics for Energy Aware Smart Campus *(Indian Institute of Science)*

- Analyzed grid signals in real time to determine accurate frequency and phase at a high sampling rate.
- Developed an adaptive algorithm that can accurately estimate frequency of grid signals.

Real Time Sign Language Processing System *(M.Sc. Computer Science Project, St. Xavier's College)*

- Developed a system to recognize the American Sign Language and convert it to English Language in real time using a perceptron-based network.
- Extended the interface on to include Braille.

Refereed Publications

- **Ghosh, A.**, Cheema, N., Oguz, C., Theobalt, C., Slusallek, P. Synthesis of Compositional Animations from Textual Descriptions. Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV) 2021 (pp 1396-1406). *[Poster]*
- **Ghosh, A.**, Cheema, N., Oguz, C., Theobalt, C., Slusallek, P. Text-Based Motion Synthesis with a Hierarchical Two-Stream RNN, SIGGRAPH 2021. *[Poster]*
- Bansal, S., **Ghosh, A.**, Seelamantula, C. S., Gurrala, G., Ghosh, P. K. Adaptive Frequency Estimation Using Iterative DESA with RDFT-based Filter. In Asia-Pacific Power and Energy Engineering Conference (APPEEC), 2017 IEEE PES (pp. 1-6). IEEE. *[Oral]*
- Seth, D., **Ghosh, A.**, Dasgupta, A., Nath, A. Real Time Sign Language Processing System. In 2016 International Conference on Smart Trends for Information Technology and Computer Communications (pp. 11-18). Springer, Singapore. *[Oral]*
- Dasgupta, A., Seth, D., **Ghosh, A.**, Nath, A. Real Time Sign Language to Braille Interfacing System. In 2017 7th International Conference on Communication Systems and Network Technologies (CSNT) (pp. 371-375). IEEE. *[Oral]*

Awards

- **Best Master's Project Award 2016.** From St. Xavier's College, Kolkata, India, for the project "Real-Time Sign Language Processing System".
- **Certificate of Merit 2013.** From St. Xavier's College, Kolkata, India, for securing the first position in semesters III and IV of Bachelor of Science in Computer Science.
- **Vidyasagar Scholarship 2009.** From the Government of West Bengal, India, for securing second rank among girls in Kolkata in the statewide 10th standard examination.

Professional Service

- **Conference Volunteer** at the IEEE PES Asia-Pacific Power and Energy Engineering Conference (APPEEC), 2017.